

Gap Analysis: `Physics::Ellipsometry::VASE` vs. `refellips`

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1 Executive Summary

`Physics::Ellipsometry::VASE` is a Perl/PDL-based generic curve-fitting framework for spectroscopic ellipsometry data, while `refellips` (Python/NumPy) is a purpose-built ellipsometry analysis package with built-in optical models, transfer matrix methods, and dispersion relations.

After extensive debugging and patching, both packages successfully fitted the Cap_01242007 Ta₂O₅/Ta metal dataset (6 wafers, 1887 points each) with comparable MSE values (0.87–0.92). However, achieving this with VASE required approximately 300 lines of hand-coded physics in the user script, whereas `refellips` required approximately 40 lines of configuration.

Table 1 compares the fitting results from both packages.

Table 1: Comparison of fitting results: VASE (Perl) vs. `refellips` (Python)

Wafer	Ta ₂ O ₅ (nm)		MSE	
	VASE	<code>refellips</code>	VASE	<code>refellips</code>
1	216.3	215.9	0.896	0.892
2	216.1	216.1	0.890	0.883
3	216.7	216.5	0.898	0.894
4	181.9	181.4	0.874	0.673
5	183.1	182.1	0.918	0.784
6	181.7	181.2	0.890	0.679

2 Architecture Comparison

2.1 `refellips`

- Transfer matrix method (TMM) adapted from Byrnes’ `tmm` package
- Built-in dispersion models: Cauchy, Sellmeier, Tauc-Lorentz, Genosc
- Built-in EMA mixing: Linear, Bruggeman, Maxwell-Garnett
- Material file (`.mat`) loader for point-by-point optical constants
- `ObjectiveSE` with circular Delta residuals and delta convention handling
- Parameter bounds and vary/fix control via `refnx.analysis.Parameter`
- Differential evolution global optimizer + Levenberg-Marquardt refinement

2.2 Physics::Ellipsometry::VASE

- Generic Levenberg-Marquardt fitting wrapper around `PDL::Fit::LM`
- Woollam VASE data file loader (4-line header, 6-column format)
- Gnuplot-based plotting of Psi/Delta data with model overlay
- **No built-in optical models**—user must code all physics
- **No TMM, Fresnel equations, dispersion, EMA, or material loaders**

3 Identified Gaps

The following gaps were identified during the process of porting the `fit_refellips.py` script to Perl using the VASE module. Each gap is listed with severity (Critical/Major/Minor) and the workaround used.

3.1 Critical Gaps

3.1.1 No Transfer Matrix Method (TMM)

Impact: Users must hand-code Fresnel equations, Snell’s law, phase accumulation, and multi-layer back-propagation for every model. This required ~ 80 lines of complex-arithmetic PDL code in `fit_vase.pl`.

Workaround: Implemented nested Airy formula (back-propagation) directly in the model function closure. During debugging, discovered that the phase convention must use $e^{+2i\beta}$ (physics convention), not $e^{-2i\beta}$ (engineering convention).

3.1.2 No Delta Convention Handling

Impact: Ellipsometry instruments (e.g., Woollam WVASE) output Delta in various ranges ($[-90^\circ, 270^\circ]$ or $[0^\circ, 360^\circ]$), while the complex argument $\arg(\rho)$ gives $(-180^\circ, 180^\circ]$. Without proper handling, LM residuals can be $\sim 360^\circ$ at wrap boundaries.

refellips solution:

$$\Delta = \angle\left(\frac{1}{-\rho}\right) + \pi = -\arg(\rho) \bmod 2\pi \quad (1)$$

combined with circular distance residuals: $r = \min(|\Delta_d - \Delta_m|, 360 - |\Delta_d - \Delta_m|)$.

Workaround: Implemented $-\arg(\rho)$ mapped to $[0^\circ, 360^\circ)$ in the model function, plus alignment to data values via `rint(diff/360)` rounding to eliminate wrap discontinuities.

3.1.3 Singular Jacobian in LM Regularization

Impact: `PDL::Fit::LM` regularizes the normal equations as $\text{diag} * (1 + \lambda)$. When any Jacobian column is zero (common with clipped or insensitive parameters), the diagonal stays zero and LU decomposition fails with “singular matrix.”

refellips solution: Uses `scipy.optimize.least_squares` with trust-region-reflective algorithm, which handles singular Jacobians via SVD internally.

Workaround: Patched `PDL::Fit::LM` line 122 to add $10^{-10}\lambda$ to the diagonal after scaling, and line 158 to regularize the covariance matrix. Also added configurable `deriv_step` and `min_deriv_step` to `VASE.pm`.

3.2 Major Gaps

3.2.1 No Dispersion Models

Impact: No built-in Cauchy, Sellmeier, Tauc-Lorentz, Drude, or General Oscillator (Genosc) models. Each must be coded manually.

Workaround: Implemented Cauchy dispersion ($n = A + B/\lambda^2 + C/\lambda^4$) directly in the model function (3 lines of code).

3.2.2 No EMA Mixing Rules

Impact: No Bruggeman, Maxwell-Garnett, or linear effective medium approximation. Must be coded by hand.

Workaround: Implemented linear volume-weighted mixing ($\varepsilon_{\text{eff}} = (1 - f)\varepsilon_A + f\varepsilon_B$) in the model function. Note: refellips defaults to linear mixing as well, but provides Bruggeman and Maxwell-Garnett options.

3.2.3 No Material File Loader

Impact: Point-by-point optical constants (.mat files with wavelength/n/k columns) must be loaded manually with custom parsing and interpolation code.

Workaround: Wrote a `load_mat_file()` function (30 lines) with eV/nm unit handling and PDL interpolation.

3.2.4 No Parameter Bounds or Vary/Fix Control

Impact: Parameters cannot be constrained to physical ranges (e.g., thickness > 0 , $0 \leq f_v \leq 1$). All parameters are always varied. Unphysical parameter values can crash the model.

Workaround: Used `abs()` for thickness and `clip(0.01, 0.999)` for volume fraction inside the model function. This creates discontinuous derivatives at the boundaries.

3.2.5 No Global Optimizer

Impact: LM is a local optimizer. Without a global search (differential evolution, genetic algorithm), the fit is sensitive to initial parameter guesses and can get trapped in local minima.

Workaround: Implemented a grid search over thickness before LM, plus a second-pass refit using neighbor wafer parameters for wafers that converge to poor minima.

3.3 Minor Gaps

3.3.1 No eV $\leftrightarrow$ nm Unit Conversion

The data loader stores but does not act on the units field. Woollam files can use either nm or eV for the spectral axis.

Workaround: Added manual conversion ($\lambda_{\text{nm}} = 1239.842/E_{\text{eV}}$) and wavelength sorting after loading.

3.3.2 Windows CR Characters in Data Files

The `load_data()` method used `chomp` which removes `\n` but not `\r`. This caused the units field "eV\r" to fail string comparison.

Fix applied: Added `$units =~ s/\r//g` in `VASE.pm`.

3.3.3 NiceSlice Source Filter Ambiguity

`PDL::NiceSlice` transforms `$pdl->(...)` expressions into slice operations. The syntax `pdl[$var]` and `$coderef->(...)` are incorrectly parsed as slices, producing garbage values silently.

Workaround: Use `pdl($var)` (explicit parens) and `&$coderef(...)` syntax in all code using `NiceSlice`.

3.3.4 Hardcoded Finite-Difference Step

The original derivative step ($|\mathbf{p}_i| \times 10^{-7} + 10^{-10}$) was too small for thick-film models where ~ 2000 Å thickness parameters need $\sim 0.1\%$ perturbations.

Fix applied: Made `deriv_step` and `min_deriv_step` configurable in `VASE.pm`.

4 Recommended Improvements

In priority order:

1. **Built-in TMM engine** with proper phase conventions ($e^{+2i\beta}$, Verdet sign for r_p).
2. **Delta convention handling** with circular distance residuals in the fitting objective function.
3. **LM regularization fix:** Use `diag +=  $\lambda$` (Marquardt) instead of `diag *= (1 +  $\lambda$ )` (Levenberg), or switch to SVD-based solver.
4. **Dispersion model library:** Cauchy, Sellmeier, Tauc-Lorentz, Drude, General Oscillator.
5. **EMA mixing rules:** Linear, Bruggeman, Maxwell-Garnett.
6. **Material file loader** with automatic eV/nm conversion.
7. **Parameter bounds and vary/fix control** (e.g., via log-barrier or transformation methods).
8. **Global optimizer** (differential evolution or simulated annealing) for initial parameter estimation.

5 Files Modified

`lib/Physics/Ellipsometry/VASE.pm` — Added configurable `deriv_step`, `min_deriv_step`; fixed Windows CR in header parsing.

`PDL/Fit/LM.pm` — Added diagonal regularization floor ($+10^{-10}\lambda$) to prevent singular matrices; regularized covariance matrix computation.

`examples/Cap_01242007/fit_vase.pl` — Created: full ellipsometry model implementation (TMM, Cauchy, EMA, material loading, Delta conventions) using VASE as the fitting backend.

6 Conclusion

`Physics::Ellipsometry::VASE` successfully reproduced the `refellips` fitting results for all 6 wafers in the `Cap_01242007` dataset, achieving MSE values of 0.87–0.92 compared to `refellips`’ 0.67–0.89. The slightly higher MSE for wafers 4–6 (0.87–0.92 vs. 0.67–0.78) may be due to differences in the optimizer (LM vs. differential evolution + LM) or the linear vs. Bruggeman EMA choice.

However, the effort required was substantially greater: the user must implement all optical physics from scratch, handle Delta conventions manually, and work around several bugs in the LM implementation. The 8 recommended improvements would transform VASE from a generic fitting wrapper into a competitive ellipsometry analysis package.